

GOOGLE ADS SALES DATA ANALYSIS

Name: Meghna Manjesh

1. Dataset Used

For this project, I used the Google Ads Sales dataset from Kaggle. The dataset had 2,600 records and different columns related to advertisements, such as clicks, impressions, cost, leads, conversions, sales, date, location, device, and keyword.

Dataset link: <https://www.kaggle.com/datasets/nayakganes007/google-ads-sales-dataset>

2. Opened the Raw Dataset

First, I opened the downloaded dataset in Excel. I kept the original data separately in the Raw Data sheet so that I would not change the original dataset while doing the cleaning.

3. Converted the Data into an Excel Table

I selected the complete dataset and converted it into an Excel Table. This made it easier to filter the columns, work with the records and manage the data during the analysis.

Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword
A1000	DataAnalyticsCourse	104	4498	\$231.88	14	7	0.058	\$1,892	11/16/2024	hyderabad	desktop	learn data analytics
A1001	DataAnalyticsCourse	173	5107	\$216.84	10	8	0.046	\$1,679	20-11-2024	hyderabad	mobile	data analytics course
A1002	Data Analytics Corse	90	4544	\$203.66	26	9		\$1,624	11/16/2024	hyderabad	Desktop	data analytics online
A1003	Data Analytcis Course	142	3185	\$237.66	17	6		\$1,225	11/26/2024	HYDERABAD	tablet	data analytics training
A1004	Data Analytics Corse	156	3361	\$195.90	30	8		\$1,091	11/22/2024	hyderabad	desktop	online data analytic
A1005	DataAnalyticsCourse	195	3776	\$243.57	10	8		\$1,315	16-11-2024	hyderabad	MOBILE	data analytics training
A1006	Data Analytics Corse	116	4480	\$237.79	17	5	0.043	\$1,640	6/11/2024	hyderabad	TABLET	data analytics course
A1007	Data Analytics Corse	184	5060	\$229.61	23	3	0.016	\$1,509	11/24/2024	Hyderabad	TABLET	analytics for data
A1008	Data Analytics Corse	113	5434		27	4	0.058	\$1,362	11/24/2024	Hyderabad	Tablet	data analytics training
A1009	Data Analytics Course	166	3355	\$186.78	24	9	0.054	\$1,029	11/12/2024	Hyderabad	Mobile	online data analytic
A1010	DataAnalyticsCourse	101	5399	\$236.79	20	6	0.059	\$1,900	14-11-2024	HYDERABAD	DESKTOP	learn data analytics
A1011	Data Analytics Corse	101	3613	\$208.12	24	5	0.05	\$1,130	11/22/2024	hyderabad	Desktop	online data analytic
A1012	Data Analytics Course	125	3259	\$191.30	18	3	0.024	\$1,959	11/27/2024	hyderabad	DESKTOP	data analytics online
A1013	Data Analytics Corse	196	3742	\$207.46	10	7	0.038	\$1,623	12/11/2024	hyderabad	mobile	data analytics training
A1014	Data Analytcis Course	181	4311	\$185.09	11	9	0.05	\$1,538	11/4/2024	Hyderabad	Desktop	data analytics training
A1015	Data Analytics Course	102	5461	\$201.69	22	8	0.078	\$1,755	11/29/2024	hyderabad	desktop	data analytics course
A1016	Data Analytcis Course	193	5159		15	9	0.047	\$1,614	11/10/2024	hydrebad	Mobile	learn data analytics
A1017	Data Analytics Corse	149	4431	\$207.40	19	9	0.06	\$1,693	12/11/2024	HYDERABAD	mobile	learn data analytics
A1018	Data Analytics Corse	150	3113	\$217.41	20	5		\$1,785	11/7/2024	hyderabad	Mobile	analytics for data
A1019	Data Analytics Course	145	5278		25	6	0.041	\$1,516	11/5/2024	hyderabad	DESKTOP	data analytics course

4. Checked for Duplicate Records

I checked the Ad_ID column for duplicate values. The purpose was to make sure that the same advertisement record was not repeated. No duplicate Ad_ID records were found.

The screenshot shows a Microsoft Excel spreadsheet titled 'GoogleAds_DataAnalytics_Sales_Unc...'. The data is organized in a table with columns: Ad_ID, Campaign_Name, Clicks, Impressions, Cost, Leads, Conversions, Conversion Rate, Sale_Amount, Ad_Date, Location, Device, and Keyword. The table contains 21 rows of data. A message box from Excel is displayed in the center, stating 'No duplicate values found.' with an 'OK' button. The message box also includes a link to 'Give Feedback to Microsoft'.

Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword
A1000	DataAnalyticsCourse	104	4498	\$231.88	14	7	0.058	\$1,892	11/16/2024	hyderabad	desktop	learn data analytics
A1001	DataAnalyticsCourse	173	5107	\$216.84	10	8	0.046	\$1,679	20-11-2024	hyderabad	mobile	data analytics course
A1002	Data Analytics Corse	90	4544	\$203.66				\$1,624	11/16/2024	hyderabad	Desktop	data analytics online
A1003	Data Analytcis Course	142	3185	\$237.66				\$1,225	11/26/2024	HYDERABAD	tablet	data anaytics training
A1004	Data Analytics Corse	156	3361	\$195.90				\$1,091	11/22/2024	hyderabad	desktop	online data analytic
A1005	DataAnalyticsCourse	195	3776	\$243.57				\$1,315	16-11-2024	hyderabad	MOBILE	data anaytics training
A1006	Data Analytics Corse	116	4480	\$237.79				\$1,640	6/11/2024	hyderabad	TABLET	data analytics course
A1007	Data Analytics Corse	184	5060	\$229.61				\$1,509	11/24/2024	Hyderabad	TABLET	analytics for data
A1008	Data Analytics Corse	113	5434					\$1,362	11/24/2024	Hyderabad	Tablet	data analytics training
A1009	Data Analytcis Course	166	3355	\$186.78				\$1,029	11/12/2024	Hyderabad	Mobile	online data analytic
A1010	DataAnalyticsCourse	101	5399	\$236.79				\$1,900	14-11-2024	HYDERABAD	DESKTOP	learn data analytics
A1011	Data Analytics Corse	101	3613	\$208.12	24	5	0.05	\$1,130	11/22/2024	hyderabad	Desktop	online data analytic
A1012	Data Analytcis Course	125	3259	\$191.30	18	3	0.024	\$1,959	11/27/2024	hydrebad	DESKTOP	data analytics online
A1013	Data Analytics Corse	196	3742	\$207.46	10	7	0.038	\$1,623	12/11/2024	hyderabad	mobile	data analytics training
A1014	Data Analytics Corse	181	4311	\$185.09	11	9	0.05	\$1,538	11/4/2024	Hyderabad	Desktop	data analytics training
A1015	Data Analytcis Course	102	5461	\$201.69	22	8	0.078	\$1,755	11/29/2024	hyderabad	desktop	data analytics course
A1016	Data Analytcis Course	193	5159		15	9	0.047	\$1,614	11/10/2024	hydrebad	Mobile	learn data analytics
A1017	Data Analytics Corse	149	4431	\$207.40	19	9	0.06	\$1,693	12/11/2024	HYDERABAD	mobile	learn data analytics
A1018	Data Analytics Corse	150	3113	\$217.41	20	5		\$1,785	11/7/2024	hyderabad	Mobile	analytics for data
A1019	Data Analytcis Course	145	5278		25	6	0.041	\$1,516	11/5/2024	hyderabad	DESKTOP	data analytics course

5. Cleaned Campaign Names

The Campaign_Name column had different spellings for the same campaign. For example, 'Data Analytcis Course', 'Data Analytics Corse', 'DataAnalyticsCourse' and 'Data Analytics Corse' were present. I standardized these values to 'Data Analytics Course'.

onedrive.live.com/personal/63AD3FC7FF49F05F/_layouts/15/doc2.aspx?sourcedoc=%7Bcd3105ae-c6a2-47dc-8c89-e6dd111f6902%7D&act...

GoogleAds_DataAnalytics_Sales_Unc...

Search for tools, help, and more (Alt + Q)

Buy Microsoft 365

File Home Insert Share Page Layout Formulas Data Review View Help Draw Table Design

Comments Catch up Editing Share

Calibri (Body) 11 B

General

B2 Data Analytics Course

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword	
2	A1000	Data Analytics Course	104	4498	\$231.88	14	7	0.058	\$1,892	11/16/2024	hyderabad	desktop	learn data analytics	
3	A1001	Data Analytics Course	173	5107	\$216.84	10	8	0.046	\$1,679	20-11-2024	hyderabad	mobile	data analytics course	
4	A1002	Data Analytics Course	90	4544	\$203.66	26	9		\$1,624	11/16/2024	hyderabad	Desktop	data analytics online	
5	A1003	Data Analytics Course	142	3185	\$237.66	17	6		\$1,225	11/26/2024	HYDERABAD	tablet	data analytics training	
6	A1004	Data Analytics Course	156	3361	\$195.90	30	8		\$1,091	11/22/2024	hyderabad	desktop	online data analytic	
7	A1005	Data Analytics Course	195	3776	\$243.57	10	8		\$1,315	16-11-2024	hyderabad	MOBILE	data analytics training	
8	A1006	Data Analytics Course	116	4480	\$237.79	17	5	0.043	\$1,640	6/11/2024	hyderabad	TABLET	data analytics course	
9	A1007	Data Analytics Course	184	5060	\$229.61	23	3	0.016	\$1,509	11/24/2024	Hyderabad	TABLET	analytics for data	
10	A1008	Data Analytics Course	113	5434		27	4	0.058	\$1,362	11/24/2024	Hyderabad	Tablet	data analytics training	
11	A1009	Data Analytics Course	166	3355	\$186.78	24	9	0.054	\$1,029	11/12/2024	Hyderabad	Mobile	online data analytic	
12	A1010	Data Analytics Course	101	5399	\$236.79	20	6	0.059	\$1,900	14-11-2024	HYDERABAD	DESKTOP	learn data analytics	
13	A1011	Data Analytics Course	101	3613	\$208.12	24	5	0.05	\$1,130	11/22/2024	hyderabad	Desktop	online data analytic	
14	A1012	Data Analytics Course	125	3259	\$191.30	18	3	0.024	\$1,959	11/27/2024	hydrebad	DESKTOP	data analytics online	
15	A1013	Data Analytics Course	196	3742	\$207.46	10	7	0.038	\$1,623	12/11/2024	hyderabad	mobile	data analytics training	
16	A1014	Data Analytics Course	181	4311	\$185.09	11	9	0.05	\$1,538	11/4/2024	Hyderabad	Desktop	data analytics training	
17	A1015	Data Analytics Course	102	5461	\$201.69	22	8	0.078	\$1,755	11/29/2024	hyderabad	desktop	data analytics course	
18	A1016	Data Analytics Course	193	5159		15	9	0.047	\$1,614	11/10/2024	hydrebad	Mobile	learn data analytics	
19	A1017	Data Analytics Course	149	4431	\$207.40	19	9	0.06	\$1,693	12/11/2024	HYDERABAD	mobile	learn data analytics	
20	A1018	Data Analytics Course	150	3113	\$217.41	20	5		\$1,785	11/7/2024	hyderabad	Mobile	analytics for data	
21	A1019	Data Analytics Course	145	5278		25	6	0.041	\$1,516	11/5/2024	hyderabad	DESKTOP	data analytics course	

Workbook Statistics

Count: 2600

100%

19:39 30-09-2026

6. Cleaned Location Names

The Location column also had inconsistent spellings and capitalization. Values such as 'hyderabad', 'HYDERABAD', 'Hyderabad' and 'hydrebad' were standardized to 'Hyderabad'.

onedrive.live.com/personal/63AD3FC7FF49F05F/_layouts/15/doc2.aspx?sourcedoc=%7Bcd3105ae-c6a2-47dc-8c89-e6dd111f6902%7D&act...

GoogleAds_DataAnalytics_Sales_Unc...

Search for tools, help, and more (Alt + Q)

Buy Microsoft 365

File Home Insert Share Page Layout Formulas Data Review View Help Draw Table Design

Comments Catch up Editing Share

Calibri (Body) 11 B General \$€ ¥ .00 20%

K2 Hyderabad

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword	
2	A1000	Data Analytics Course	104	4498	\$231.88	14	7	0.058	\$1,892	11/16/2024	Hyderabad	desktop	learn data analytics	
3	A1001	Data Analytics Course	173	5107	\$216.84	10	8	0.046	\$1,679	20-11-2024	Hyderabad	mobile	data analytics course	
4	A1002	Data Analytics Course	90	4544	\$203.66	26	9		\$1,624	11/16/2024	Hyderabad	Desktop	data analytics online	
5	A1003	Data Analytics Course	142	3185	\$237.66	17	6		\$1,225	11/26/2024	Hyderabad	tablet	data analytics training	
6	A1004	Data Analytics Course	156	3361	\$195.90	30	8		\$1,091	11/22/2024	Hyderabad	desktop	online data analytic	
7	A1005	Data Analytics Course	195	3776	\$243.57	10	8		\$1,315	16-11-2024	Hyderabad	MOBILE	data analytics training	
8	A1006	Data Analytics Course	116	4480	\$237.79	17	5	0.043	\$1,640	6/11/2024	Hyderabad	TABLET	data analytics course	
9	A1007	Data Analytics Course	184	5060	\$229.61	23	3	0.016	\$1,509	11/24/2024	Hyderabad	TABLET	analytics for data	
10	A1008	Data Analytics Course	113	5434		27	4	0.058	\$1,362	11/24/2024	Hyderabad	Tablet	data analytics training	
11	A1009	Data Analytics Course	166	3355	\$186.78	24	9	0.054	\$1,029	11/12/2024	Hyderabad	Mobile	online data analytic	
12	A1010	Data Analytics Course	101	5399	\$236.79	20	6	0.059	\$1,900	14-11-2024	Hyderabad	DESKTOP	learn data analytics	
13	A1011	Data Analytics Course	101	3613	\$208.12	24	5	0.05	\$1,130	11/22/2024	Hyderabad	Desktop	online data analytic	
14	A1012	Data Analytics Course	125	3259	\$191.30	18	3	0.024	\$1,959	11/27/2024	Hyderabad	DESKTOP	data analytics online	
15	A1013	Data Analytics Course	196	3742	\$207.46	10	7	0.038	\$1,623	12/11/2024	Hyderabad	mobile	data analytics training	
16	A1014	Data Analytics Course	181	4311	\$185.09	11	9	0.05	\$1,538	11/4/2024	Hyderabad	Desktop	data analytics training	
17	A1015	Data Analytics Course	102	5461	\$201.69	22	8	0.078	\$1,755	11/29/2024	Hyderabad	desktop	data analytics course	
18	A1016	Data Analytics Course	193	5159		15	9	0.047	\$1,614	11/10/2024	Hyderabad	Mobile	learn data analytics	
19	A1017	Data Analytics Course	149	4431	\$207.40	19	9	0.06	\$1,693	12/11/2024	Hyderabad	mobile	learn data analytics	
20	A1018	Data Analytics Course	150	3113	\$217.41	20	5		\$1,785	11/7/2024	Hyderabad	Mobile	analytics for data	
21	A1019	Data Analytics Course	145	5278		25	6	0.041	\$1,516	11/5/2024	Hyderabad	DESKTOP	data analytics course	

Workbook Statistics

Count: 2600 100%

19:42 30-09-2026

7. Standardized Device Names

The Device column contained the same device in different cases, such as Desktop, desktop, and DESKTOP. I standardized the values into three consistent categories: Desktop, Mobile, and Tablet.

onedrive.live.com/personal/63AD3FC7FF49F05F/_layouts/15/doc2.aspx?sourcedoc=%7Bcd3105ae-c6a2-47dc-8c89-e6dd111f6902%7D&act...

GoogleAds_DataAnalytics_Sales_Unc...

Search for tools, help, and more (Alt + Q)

Buy Microsoft 365

File Home Insert Share Page Layout Formulas Data Review View Help Draw Table Design

Comments Catch up Editing Share

Calibri (Body) 11 B

General

Desktop

Find and Replace

Find Replace

Find

Replace with

Search options

Matches replaced (834)

Find next Find all Replace Replace all

Ad_ID	Campaign_Name	Clicks	Impressions
A1000	Data Analytics Course	104	4498
A1001	Data Analytics Course	173	5107
A1002	Data Analytics Course	90	4544
A1003	Data Analytics Course	142	3185
A1004	Data Analytics Course	156	3361
A1005	Data Analytics Course	195	3776
A1006	Data Analytics Course	116	4480
A1007	Data Analytics Course	184	5060
A1008	Data Analytics Course	113	5434
A1009	Data Analytics Course	166	3355
A1010	Data Analytics Course	101	5399
A1011	Data Analytics Course	101	3613
A1012	Data Analytics Course	125	3259
A1013	Data Analytics Course	196	3742
A1014	Data Analytics Course	181	4311
A1015	Data Analytics Course	102	5461
A1016	Data Analytics Course	193	5159
A1017	Data Analytics Course	149	4431
A1018	Data Analytics Course	150	3113
A1019	Data Analytics Course	145	5278

Date	Location	Device	Keyword
16/2024	Hyderabad	Desktop	learn data analytics
1-2024	Hyderabad	Mobile	data analytics course
16/2024	Hyderabad	Desktop	data analytics online
26/2024	Hyderabad	Tablet	data analytics training
22/2024	Hyderabad	Desktop	online data analytic
1-2024	Hyderabad	Mobile	data analytics training
11/2024	Hyderabad	Tablet	data analytics course
24/2024	Hyderabad	Tablet	analytics for data
24/2024	Hyderabad	Tablet	data analytics training
12/2024	Hyderabad	Mobile	online data analytic
1-2024	Hyderabad	Desktop	learn data analytics
22/2024	Hyderabad	Desktop	online data analytic
27/2024	Hyderabad	Desktop	data analytics online
11/2024	Hyderabad	Mobile	data analytics training
7/4/2024	Hyderabad	Desktop	data analytics training
29/2024	Hyderabad	Desktop	data analytics course
10/2024	Hyderabad	Mobile	learn data analytics
12/11/2024	Hyderabad	Mobile	learn data analytics
11/7/2024	Hyderabad	Mobile	analytics for data
11/5/2024	Hyderabad	Desktop	data analytics course

Workbook Statistics

Count: 2600

100%

19:45 30-09-2026

8. Cleaned Cost and Sale Amount

Cost and Sale Amount contained values with currency symbols or text formatting. I cleaned and formatted these columns as numbers so that they could be used correctly in calculations.

onedrive.live.com/personal/63AD3FC7FF49F05F/_layouts/15/doc2.aspx?sourcedoc=%7Bcd3105ae-c6a2-47dc-8c89-e6dd111f6902%7D&action=e...

GoogleAds_DataAnalytics_Sales_Un...

Search for tools, help, and more (Alt + Q)

Buy Microsoft 365

File Home Insert Share Page Layout Formulas Data Review View Help Draw Table Design

Comments Catch up Editing Share

Calibri Bold 11 B

Number

fx Sale_Amount

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword	
2	A1000	Data Analytics Course	104	4498	231.88	14	7	0.058	1892.00	11/16/2024	Hyderabad	Desktop	learn data analytics	
3	A1001	Data Analytics Course	173	5107	216.84	10	8	0.046	1679.00	20-11-2024	Hyderabad	Mobile	data analytics course	
4	A1002	Data Analytics Course	90	4544	203.66	26	9		1624.00	11/16/2024	Hyderabad	Desktop	data analytics online	
5	A1003	Data Analytics Course	142	3185	237.66	17	6		1225.00	11/26/2024	Hyderabad	Tablet	data analytics training	
6	A1004	Data Analytics Course	156	3361	195.90	30	8		1091.00	11/22/2024	Hyderabad	Desktop	online data analytic	
7	A1005	Data Analytics Course	195	3776	243.57	10	8		1315.00	16-11-2024	Hyderabad	Mobile	data analytics training	
8	A1006	Data Analytics Course	116	4480	237.79	17	5	0.043	1640.00	6/11/2024	Hyderabad	Tablet	data analytics course	
9	A1007	Data Analytics Course	184	5060	229.61	23	3	0.016	1509.00	11/24/2024	Hyderabad	Tablet	analytics for data	
10	A1008	Data Analytics Course	113	5434		27	4	0.058	1362.00	11/24/2024	Hyderabad	Tablet	data analytics training	
11	A1009	Data Analytics Course	166	3355	186.78	24	9	0.054	1029.00	11/12/2024	Hyderabad	Mobile	online data analytic	
12	A1010	Data Analytics Course	101	5399	236.79	20	6	0.059	1900.00	14-11-2024	Hyderabad	Desktop	learn data analytics	
13	A1011	Data Analytics Course	101	3613	208.12	24	5	0.05	1130.00	11/22/2024	Hyderabad	Desktop	online data analytic	
14	A1012	Data Analytics Course	125	3259	191.30	18	3	0.024	1959.00	11/27/2024	Hyderabad	Desktop	data analytics online	
15	A1013	Data Analytics Course	196	3742	207.46	10	7	0.038	1623.00	12/11/2024	Hyderabad	Mobile	data analytics training	
16	A1014	Data Analytics Course	181	4311	185.09	11	9	0.05	1538.00	11/4/2024	Hyderabad	Desktop	data analytics training	
17	A1015	Data Analytics Course	102	5461	201.69	22	8	0.078	1755.00	11/29/2024	Hyderabad	Desktop	data analytics course	
18	A1016	Data Analytics Course	193	5159		15	9	0.047	1614.00	11/10/2024	Hyderabad	Mobile	learn data analytics	
19	A1017	Data Analytics Course	149	4431	207.40	19	9	0.06	1693.00	12/11/2024	Hyderabad	Mobile	learn data analytics	
20	A1018	Data Analytics Course	150	3113	217.41	20	5		1785.00	11/7/2024	Hyderabad	Mobile	analytics for data	
21	A1019	Data Analytics Course	145	5278		25	6	0.041	1516.00	11/5/2024	Hyderabad	Desktop	data analytics course	

Workbook Statistics

Average: 1498.648111 Count: 2462 Sum: 3688173

100%

20:15 30-09-2026

9. Cleaned the Date Column

The Ad_Date column contained different date formats, for example dates written with '-' and '/'. I created a new Clean_Date column and used a formula to convert the different formats into one consistent date format.

The final date was displayed in the format dd-mmm-yyyy, for example 16-Nov-2024.

	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale Amount	Ad Date	Location	Device	Keyword	Clean Date	CTR	
1	urse	104	4498	231.88	14	7	0.058	1892.00	11/16/2024	Hyderabad	Desktop	learn data analytics	11/16/2024	
2	urse	173	5107	216.84	10	8	0.046	1679.00	20-11-2024	Hyderabad	Mobile	data analytics course	11/20/2024	
3	urse	90	4544	203.66	26	9		1624.00	11/16/2024	Hyderabad	Desktop	data analytics online	11/16/2024	
4	urse	142	3185	237.66	17	6		1225.00	11/26/2024	Hyderabad	Tablet	data analytics training	11/26/2024	
5	urse	156	3361	195.90	30	8		1091.00	11/22/2024	Hyderabad	Desktop	online data analytic	11/22/2024	
6	urse	195	3776	243.57	10	8		1315.00	16-11-2024	Hyderabad	Mobile	data analytics training	11/16/2024	
7	urse	116	4480	237.79	17	5	0.043	1640.00	6/11/2024	Hyderabad	Tablet	data analytics course	6/11/2024	
8	urse	184	5060	229.61	23	3	0.016	1509.00	11/24/2024	Hyderabad	Tablet	analytics for data	11/24/2024	
9	urse	113	5434		27	4	0.058	1362.00	11/24/2024	Hyderabad	Tablet	data analytics training	11/24/2024	
10	urse	166	3355	186.78	24	9	0.054	1029.00	11/12/2024	Hyderabad	Mobile	online data analytic	11/12/2024	
11	urse	101	5399	236.79	20	6	0.059	1900.00	14-11-2024	Hyderabad	Desktop	learn data analytics	11/14/2024	
12	urse	101	3613	208.12	24	5	0.05	1130.00	11/22/2024	Hyderabad	Desktop	online data analytic	11/22/2024	
13	urse	125	3259	191.30	18	3	0.024	1959.00	11/27/2024	Hyderabad	Desktop	data analytics online	11/27/2024	
14	urse	196	3742	207.46	10	7	0.038	1623.00	12/11/2024	Hyderabad	Mobile	data analytics training	12/11/2024	
15	urse	181	4311	185.09	11	9	0.05	1538.00	11/4/2024	Hyderabad	Desktop	data analytics training	11/4/2024	
16	urse	102	5461	201.69	22	8	0.078	1755.00	11/29/2024	Hyderabad	Desktop	data analytics course	11/29/2024	
17	urse	193	5159		15	9	0.047	1614.00	11/10/2024	Hyderabad	Mobile	learn data analytics	11/10/2024	
18	urse	149	4431	207.40	19	9	0.06	1693.00	12/11/2024	Hyderabad	Mobile	learn data analytics	12/11/2024	
19	urse	150	3113	217.41	20	5		1785.00	11/7/2024	Hyderabad	Mobile	analytics for data	11/7/2024	
20	urse	145	5278		25	6	0.041	1516.00	11/5/2024	Hyderabad	Desktop	data analytics course	11/5/2024	

10. Handled Missing Values

I checked the dataset for blank values. Missing values were present in numerical columns such as Clicks, Impressions, Cost, Leads, Conversions, Sale_Amount and Conversion Rate.

Instead of deleting the records, I used the median value of each respective column to fill out the blanks. This helped me retain all 2,600 records.

	A	B	C	D	E	F	G	H	I	J	K	L
	Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device
4	A1002	Data Analytics Course	90	4544	203.66	26	9	0.046	1624.00	11/16/2024	Hyderabad	Desktop
5	A1003	Data Analytics Course	142	3185	237.66	17	6	0.046	1225.00	11/26/2024	Hyderabad	Tablet
6	A1004	Data Analytics Course	156	3361	195.90	30	8	0.046	1091.00	11/22/2024	Hyderabad	Desktop
7	A1005	Data Analytics Course	195	3776	243.57	10	8	0.046	1315.00	16-11-2024	Hyderabad	Mobile
20	A1018	Data Analytics Course	150	3113	217.41	20	5	0.046	1785.00	11/7/2024	Hyderabad	Mobile
29	A1027	Data Analytics Course	139	5286	234.89	22	10	0.046	1688.00	11/21/2024	Hyderabad	Desktop
30	A1028	Data Analytics Course	128	5469	188.49	16	3	0.046	1783.00	11/23/2024	Hyderabad	Mobile
33	A1031	Data Analytics Course	113	5065	212.37	11	5	0.046	1111.00	22-11-2024	Hyderabad	Desktop
34	A1032	Data Analytics Course	139	3872	244.38	20	4	0.046	1041.00	12/11/2024	Hyderabad	Tablet
36	A1034	Data Analytics Course	121	5610	214.96	23	8	0.046	1505.00	11/25/2024	Hyderabad	Mobile
43	A1041	Data Analytics Course	112	4271	196.07	27	6	0.046	1209.00	11/14/2024	Hyderabad	Desktop
44	A1042	Data Analytics Course	113	4418	191.55	27	10	0.046	1634.00	11/9/2024	Hyderabad	Mobile
46	A1044	Data Analytics Course	197	5783	204.46	23	6	0.046	1729.00	1/11/2024	Hyderabad	Tablet
47	A1045	Data Analytics Course	109	4866	236.78	27	5	0.046	1585.00	15-11-2024	Hyderabad	Mobile
55	A1053	Data Analytics Course	113	4945	187.66	15	10	0.046	1564.00	11/26/2024	Hyderabad	Tablet

11. Created Calculated Columns

After cleaning the data, I created a few calculated columns to analyse advertising performance.

Column
CTR
CPC
CPL
CPCONV
ROAS

I used IFERROR in these calculations so that the formula would return 0 instead of showing an error when the denominator was zero.

Metric	Mean	Median	Mode	Standard Deviation
Clicks	138.9588	139	139	33.86522752
Impressions	215.11	4518.5	4518.5	860.8434031
Cost	215.11	215.57	215.57	19.90758041
Leads	20.00385	20	20	5.97635335
Conversions	6.532692	7	7	2.241414826
Sale_Amount	1498.99	1505.00	1505	279.3272091
CTR	3.19%	3.04%	0.027027	0.0103111
Conversion Rate	0.048979	0.046	0.05	0.019987225

12. Created the Statistics Sheet

I created a separate Statistics sheet to calculate descriptive statistics for the main numerical fields. I calculated Mean, Median, Mode, and Standard Deviation for Clicks, Impressions, Cost, Leads, Conversions, Sale_Amount, CTR, and Conversion Rate.

I used AVERAGE, MEDIAN, MODE.SNGL and STDEV.S functions. The formulas were applied only to rows 2 to 2601 because the dataset contains 2,600 actual records.

13. Used SUM Function

I used the SUM function to find the total Clicks, total Cost and total Sale Amount in the dataset.

- Total Clicks = 361,293
- Total Cost = 559,282.12
- Total Sale Amount = 3,897,368

14. Used COUNTIF Function

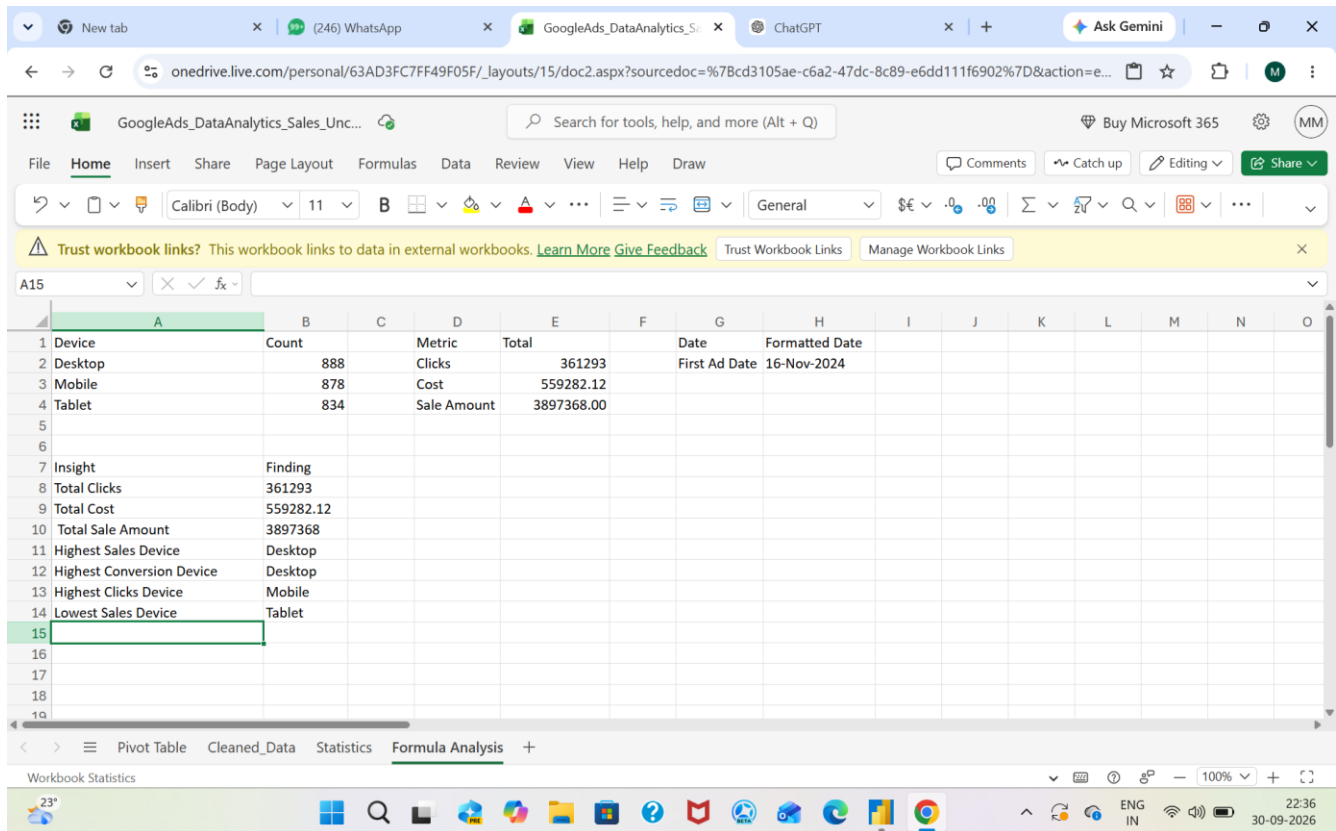
I used COUNTIF to count how many records belonged to each device category. The data range was kept fixed while copying the formula.

- Desktop = 888
- Mobile = 878

- Tablet = 834

15. Used TEXT Function

I used the TEXT function to display the cleaned date in a readable format. For example, the date was displayed as 16-Nov-2024 instead of the original mixed formats.



The screenshot shows a Microsoft Excel spreadsheet with the following data:

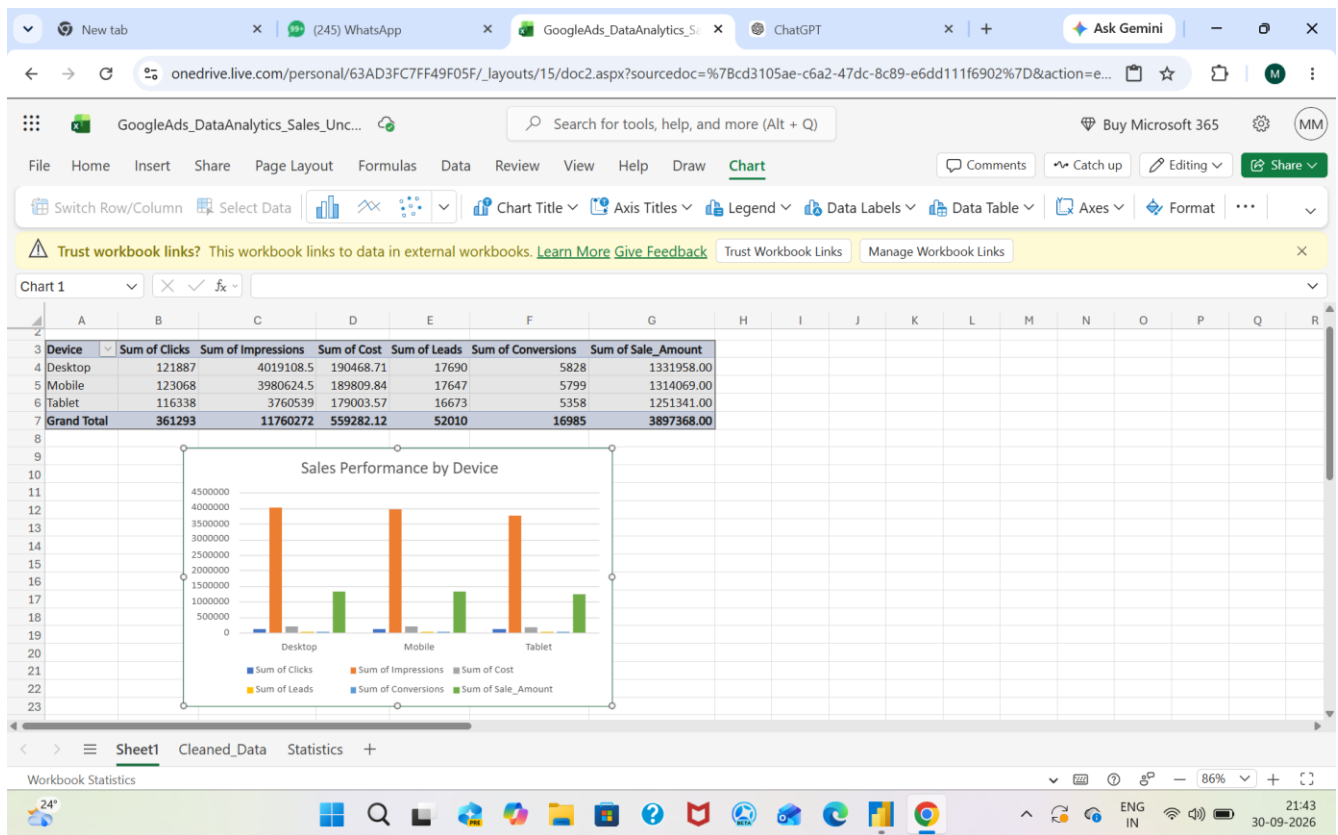
Device	Count	Metric	Total	Date	Formatted Date
Desktop	888	Clicks	361293	First Ad Date	16-Nov-2024
Mobile	878	Cost	559282.12		
Tablet	834	Sale Amount	3897368.00		
Insight	Finding				
Total Clicks	361293				
Total Cost	559282.12				
Total Sale Amount	3897368				
Highest Sales Device	Desktop				
Highest Conversion Device	Desktop				
Highest Clicks Device	Mobile				
Lowest Sales Device	Tablet				

The 'Formatted Date' column shows the date '16-Nov-2024'.

16. Created the PivotTable

After completing the cleaning and formula work, I created a PivotTable from the cleaned dataset. I used Device as the row field and added Clicks, Impressions, Cost, Leads, Conversions and Sale_Amount as value fields.

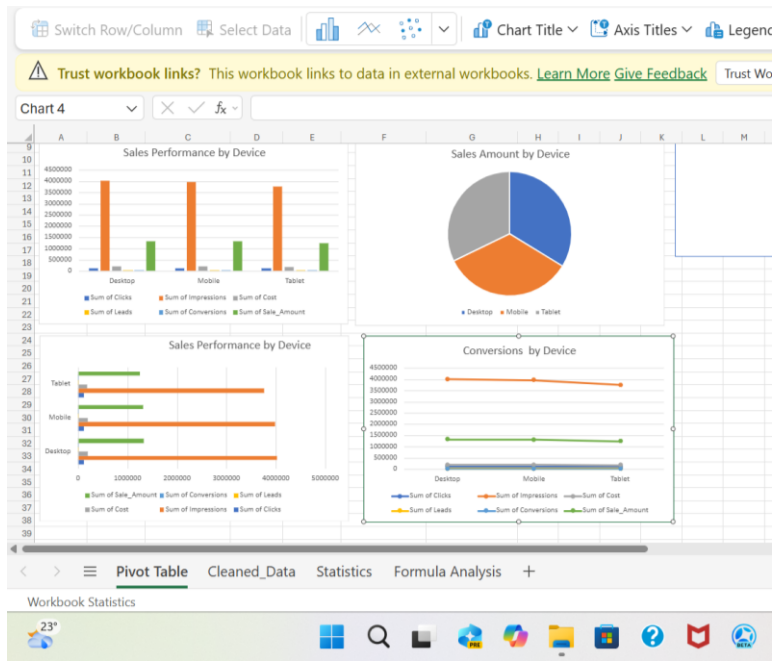
This helped me compare the advertising performance of Desktop, Mobile, and Tablet.



17. Created the Charts

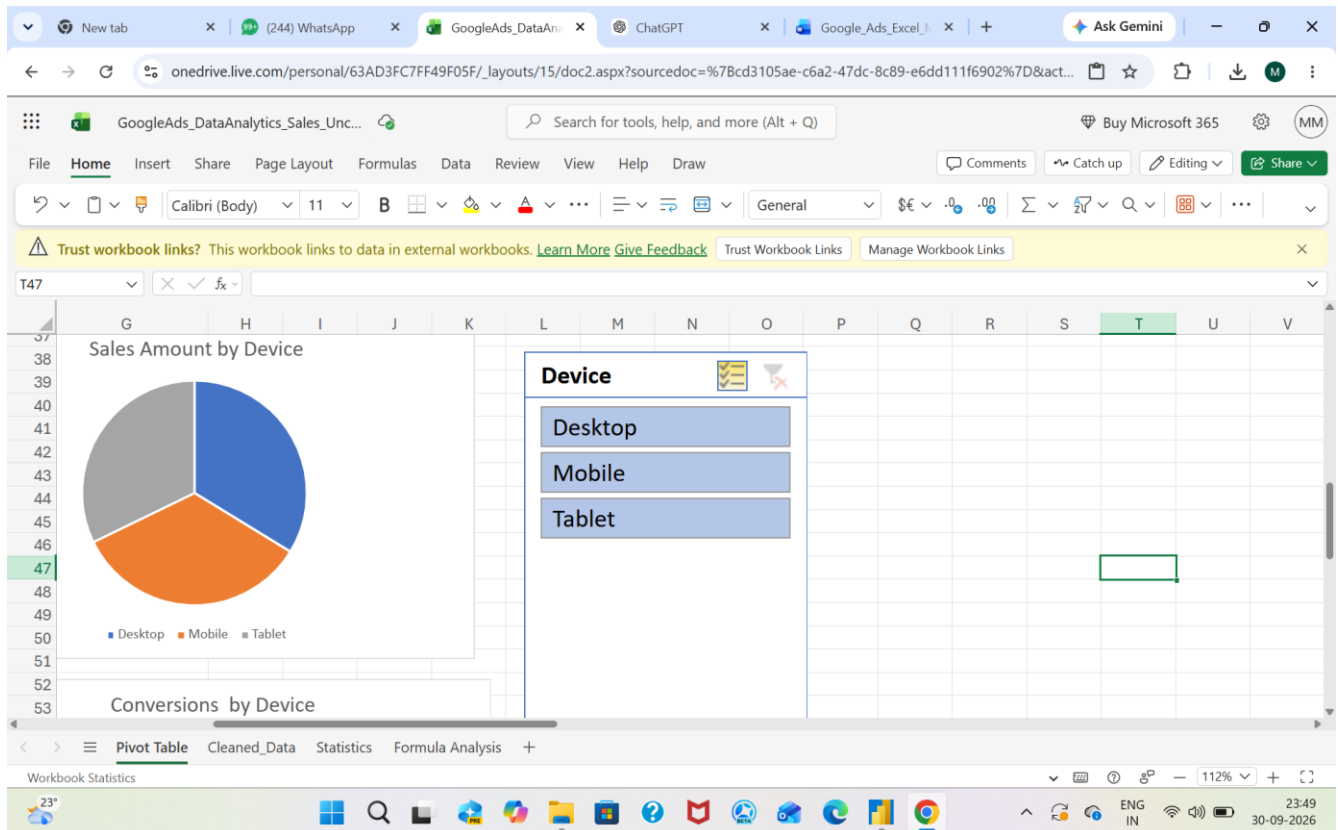
I created three different chart types from PivotTable so that the results could be understood visually.

- Clustered Column Chart – Sales Performance by Device
- Pie Chart – Sales Amount by Device
- Line Chart – Conversions by Device
- Bar Chart – Sales Performance by Device



18. Added a Device Slicer

I added a Device slicer to the dashboard. It allows the user to select Desktop, Mobile, or Tablet and view the related information interactively.



19. Final Excel Dashboard

Finally, I arranged the PivotTable, charts and slicer on Sheet 1. The purpose of the dashboard is to give a simple visual comparison of advertising performance by device.

20. Main Findings from the Excel Analysis

From the analysis, Desktop recorded the highest sales amount and conversions, while Mobile recorded the highest number of clicks. The tablet recorded the lowest sales amount among the three devices. These comparisons will be used as a starting point for the Power BI dashboard.